GITAM Deemed to be University

Electrical and Electronics Engineering | Course Code: EE304

Power Electronics & Converters - Power Semiconductor Switches & Switched-Mo

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Overview: SCR dynamic turn-on/turn-off, IGBT vs MOSFET switching losses, Buck, Boost, and Buck-

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Power Semiconductor Switching Devices
Thyristors SCR, Triac, Power MOSFET, IGBT, Switching characteristics, Safe Operating Area SOA, Gate

Unit 2: Phase-Controlled Converters
Single-phase and three-phase half-wave and full-wave controlled rectifiers with R, RL, and RLE loads

Unit 3: DC-DC Switched Mode Regulators
Buck converter, Boost converter, Buck-Boost topology, Continuous and discontinuous conduction bounda

Unit 4: Inverters & Modulation Techniques
Voltage Source Inverters VSI, Single-phase full-bridge, Three-phase 120 and 180 conduction modes, Si

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Derive the expression for the average output voltage of a single-phase fully-controlled bridge rectifi
2. Formulate the inductor volt-second balance and capacitor charge balance for a DC-DC Boost regulator in
3. Compare 180-degree and 120-degree conduction modes of a three-phase bridge inverter with line and phas

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* Ned Mohan, Tore M. Undeland, William P. Robbins - Power Electronics: Converters, Applications, and
* Muhammad H. Rashid - Power Electronics: Devices, Circuits, and Applications Pearson 4th Ed

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